Conversation: Calculating 6-Day Moving Average in R with dplyr

Question:

For a dataframe of stocks with symbol, date, year, close. How to calculate the 6 day moving average using dplyr, explain each line you will use for this to understand the idea behind it

Response:

I'll show you how to calculate a 6-day moving average using dplyr, explaining each step:

```
library(dplyr)

# Sample data structure
stocks <- data.frame(
  symbol = c("AAPL", "AAPL", "AAPL", "GOOGL", "GOOGL", "GOOGL"),
  date = as.Date(c("2024-01-01", "2024-01-02", "2024-01-03",
                   "2024-01-01", "2024-01-02", "2024-01-03")),
  year = c(2024, 2024, 2024, 2024, 2024, 2024),
  close = c(150, 152, 148, 2800, 2820, 2810)
)

# Calculate 6-day moving average
stocks_with_ma <- stocks %>%
  group_by(symbol) %>%                          # Line 1
  arrange(symbol, date) %>%                     # Line 2
  mutate(
    ma_6 = zoo::rollmean(close,                # Line 3
                         k = 6,                 # Line 4
                         fill = NA,             # Line 5
                         align = "right")       # Line 6
  ) %>%
  ungroup()                                     # Line 7
```

Explanation of each line:

Line 1: group_by(symbol)

Groups the data by stock symbol so the moving average is calculated separately for each stock. Without this, the moving average would incorrectly blend data across different stocks.

Line 2: arrange(symbol, date)

Sorts the data by symbol and then by date within each symbol. Critical because moving averages require chronologically ordered data.

Line 3: ma_6 = zoo::rollmean(close,

Creates a new column called ma_6 using the rollmean() function from the zoo package. 'close' specifies we're calculating the average of the closing prices.

Line 4: k = 6,

Sets the window size to 6 days. This means each moving average value uses 6 consecutive closing prices.

Line 5: fill = NA,

Fills positions where we don't have enough data with NA. For example, the first 5 rows won't have 6 preceding values, so they'll be NA.

Line 6: align = "right"

Aligns the window to the right (trailing average). This means each value uses the current day plus the previous 5 days. Alternative: "center" or "left" for different alignments.

Line 7: ungroup()

Removes the grouping structure. Good practice to avoid unexpected behavior in subsequent operations.

The result gives you each stock's 6-day moving average, properly calculated within each symbol group!